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Joint power minimization and trajectory design for collaborative communication, sensing, and computing in UAV networks

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ABSTRACT

In this paper, we propose a collaborative communication, sensing, and computing (CCSC) model assisted by unmanned aerial vehicles (UAVs). Ground users (GUs) generate computing tasks and communicate with UAVs to determine whether the tasks can be processed locally or offloaded to the UAVs. While GUs have limited computing power, UAVs are equipped to handle more complex tasks and sense target states. The multi-antenna base station (BS) not only manages communication tasks with UAVs but also schedules UAV operations and assigns computational tasks. The problem of UAV trajectory design and power minimization based on communication sensing and computing is studied. Since the problem is non-convex, the original problem is solved separately by a block coordinate descent approach. To solve the power minimization problem, we propose an offloading decision algorithm based on exchange matching to solve the task offloading problem and minimize the UAV computing, hovering, and flight power. For the UAV trajectory design, we propose a UAV trajectory design algorithm based on successive convex approximation (SCA) to optimize the UAV trajectory. Simulation results show that the proposed algorithm is superior to other benchmarks such as the greedy algorithm and genetic algorithm.

1. Introduction

Integrated sensing and communication (ISAC) has emerged as a fundamental component of the sixth-generation (6G) wireless networks [1]. Future 6G envisions that high-precision sensing, along with low-latency communication and computing capabilities, will transform many aspects of human life, including the Internet of Vehicles (IoV), autonomous driving, and the development of the metaverse [2]. Unmanned aerial vehicle (UAV), which is one of the most popular technologies in 6G, combined with ISAC, can improve the capacity of the communication system and benefit from enhanced sensing and communication functionalities [3]. UAVs equipped with various sensors can provide real-time environmental data, significantly boosting the efficiency of both information transmission and environmental monitoring [4]. Moreover, UAVs serve as airborne communication relays, effectively extending network coverage, which is particularly valuable in temporary disaster relief scenarios or remote areas lacking infrastructure [5]. Due to the security problem of the UAV-assisted ISAC system [6], Liu et al. proposed an ISAC–UAV assisted secure communication system that addresses the risk of eavesdroppers intercepting communication between the source UAV and users, but their focus on secrecy rates neglects the joint optimization of sensing and computing resources. This system aims to maximize the secure transmission rate while mitigating eavesdropping risks. Additionally, the authors of [7] optimized the UAV trajectory in a UAV-assisted ISAC security system and proposed a method based on Kalman filtering to maximize the real-time secrecy rates, yet their model assumes static task offloading decisions, limiting adaptability in dynamic environments.

Beamforming and UAV trajectory optimization are crucial components in UAV-assisted ISAC systems, which largely determine the performance of the system. Recently, several beamforming design methods

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have been proposed to provide better service in various UAV-assisted ISAC systems [8,9]. The authors of [8] proposed an alternate optimized beamforming method to maximize the average throughput in the system. Similarly, the authors of [9] designed trajectory and sensing beamforming vectors for UAVs, the goal is to maximize the average weighted sum rate of communication users. Trajectory design of UAV also plays an important role in the UAV–ISAC system [8–13], as an optimized trajectory significantly improves communication rates. The authors of [10] have researched an orthogonal frequency division multiplexing (OFDM) UAV–ISAC system, and put forward a joint path optimization and resource allocation problem to improve the precision of positioning. Wu et al. [11] utilized the extended Kalman filter to predict and track the motion parameters of GUs in each time slot, subsequently optimizing the UAV trajectory design using the continuous convex approximation method. The authors of [12] established a bistatic synthetic aperture radar system by combining a UAV and the base station (BS), and jointly designed the trajectory and transmission power of the UAV to minimize the cost of UAV flight and communication while meeting both sensing and communication requirements. Moreover, the authors of [13] jointly optimized the two-dimensional trajectory, speed, downlink information, and beamforming of fixed-altitude UAVs to minimize average power consumption while ensuring communication quality and sensing accuracy.

While significant progress has been made in optimizing UAV trajectories and resource allocation for communication and sensing in ISAC systems, integrating computational tasks has become increasingly relevant. Modern paradigms like aerial and edge computing now offer new opportunities to enhance the performance of UAV-assisted systems. Over-the-air computing (AirComp) has emerged as a promising technology to improve spectrum efficiency and reduce network latency by performing computations simultaneously with data transmission, seamlessly integrating communication and computation [14–17], but they often ignore the interplay between computational load and UAV mobility constraints, leading to suboptimal energy consumption. This parallel execution enhances efficiency but also presents challenges, particularly its vulnerability to interference noise, which can lead to significant computational errors [18–20]. Despite its advantages over traditional edge computing in spectrum utilization and latency, addressing these challenges remains critical for its future development. In parallel, mobile edge computing (MEC) has emerged as a complementary framework for UAV-assisted systems, offering an alternative solution by offloading computational tasks from UAVs to nearby edge servers. MEC can effectively reduce latency and preserve computing resources on the device by deploying servers in the wireless access points or BSs. As a result, MEC-enabled UAV architectures have attracted growing attention from both academia and industry, offering a feasible path toward low-latency, energy-efficient aerial intelligence. The authors of [21] investigated the computational offloading and trajectory scheduling problems of wirelessly powered MEC systems supporting UAVs and solve the long-term average energy consumption minimization problem.

Due to the integration of hardware and spectrum, the future wireless networks are shifting from providing simple services to an integrated paradigm that combines communication, sensing, and computing, and the integration of these three functions and resource coordination can achieve mutual benefit [22,23]. The authors of [24] proposed a system framework that integrates communication, sensing, and computation, focusing on waveform design and performance optimization. In [25], Huang et al. studied the UAV-assisted integrated sensing, computing, and communication (ISCC) system, where UAVs collected sensing data and determined whether to compute locally or offload tasks to edge servers, aiming to minimize the delay during resource scheduling and task assignment. To tackle the interference between the radar sensing signal and MEC, reconfigurable intelligent surfaces (RISs) have been recognized as an effective solution for enhancing system performance and minimizing energy consumption [26]. Moreover, recent integrated frameworks primarily address communication-sensing trade-offs [23–26], but fail to account for real-time task scheduling in multi-UAV networks with heterogeneous computing demands. The authors of [27] introduced an ISCC framework for UAVs, proposing a joint optimization algorithm that utilizes a semi-deterministic programming method and bump process technique to delineate the Pareto boundary between computing power and sensing beam gain. Wang et al. [28] proposed a non-orthogonal multiple access (NOMA) assisted joint communication sensing and multi-layer computing (JCSMC) framework, demonstrated significant advantages over traditional single-layer computing systems. In [29], the total power consumption of the UAV was minimized by jointly optimizing the CPU frequency, the sensing power of the UAV radar, the transmitting power, and the trajectory of the UAV, and an iterative algorithm was proposed to solve this optimization problem. The authors of [30] minimized the tracking error of ground targets by UAVs through optimizing the offloading bit sequence, the autocorrelation function, task offload amounts, and UAV location, thereby improving target tracking performance.

Building on the above research, our work differentiates from existing studies by prioritizing green energy consumption and energy-effective solutions within the integration of communication, sensing, and computing. While prior works have focused on improving performance metrics such as task offloading, waveform design, or minimizing system latency, we emphasize minimizing energy consumption through an optimized, unified collaborative communication, sensing, and computing (CCSC) framework. Collaborative communication is a core concept that refers to the close cooperation and resource sharing among UAVs, GUs, and the BS. This collaborative working model aims to improve the efficiency and performance of the entire communication network by optimizing resource allocation and task offloading. The main contributions of this work are summarized as follows:

- We propose a new CCSC model where the UAV has the function of sensing and communicating with the user, and GUs can carry out some basic computation tasks or offload the tasks to aerial servers acted by UAVs. The UAV uniformly transmits the resource occupation information to BS, then the BS centrally processes the scheduling and informs the user whether the task offloading is feasible.
- A nonconvex optimization problem is proposed to minimize the total power consumption of a collaborative communication, sensing, and computing system. The objective is to optimize the UAV's trajectory, computing task offloading, and the UAV's velocity and azimuth relative to the user.
- We use the block coordinate descent (BCD) method to decouple the problem into two sub-problems. An exchange matching-based offloading decision algorithm is introduced to minimize the computing power. In parallel, a UAV trajectory optimization algorithm leveraging the successive convex approximation (SCA) method is proposed to reduce both the hovering and flight power. Ultimately, the two algorithms work together to minimize the total power through iterative optimization within the BCD framework.
- Simulation results confirm the effectiveness of the proposed offloading decision algorithm and UAV trajectory design. When compared with the genetic and greedy algorithms, the results demonstrate that our proposed algorithm outperforms these methods, owing to its superior performance and lower computational complexity.

The rest of this article is organized as follows. Section 2 introduces the UAV-assisted CCSC model, and analyzes the power of the CCSC model and the hovering and flying power of UAVs respectively. In Section 3, we introduce the minimum total power problem, then decompose it into several sub-problems and propose solutions accordingly. In Section 4, simulation results are provided to verify that our proposed algorithm is superior to the benchmark algorithm. Section 5 is the conclusion of this paper.

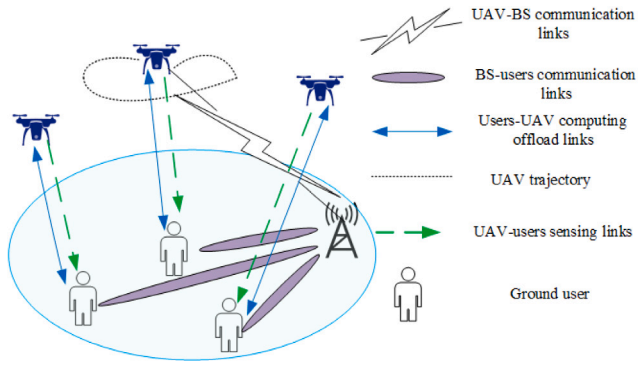

Fig. 1. A CCSC model with UAV-assisted task offloading and scheduling.

2. System model

As shown in Fig. 1, we consider a CCSC model involving UAVs operating in a town environment, which comprises M UAVs, a BS with a server, and K GUs. The UAV with the function of edge computation communicates with both the GUs and the BS while simultaneously performing the computing tasks. At the same time, the UAV can sense the user's state. The BS is used for the scheduling and task assignment of the UAV, and each user has his own computing task. Due to the limited computing resources of the user, when the UAV senses the target, it will uniformly transmit the resource occupation information to BS, and BS will centrally process the scheduling and notify the user whether the task offloading is feasible. Finally, GUs offload their computing tasks through the UAV user's communication links according to the feedback from the BS, effectively reducing the queue waiting time.

In our system, the BS serves as the unified scheduling center responsible for planning the positions, speeds of the UAVs. When UAVs sense targets, they transmit resource occupation information to the BS. Each user also reports information related to their computing tasks, such as task data volume, and required computing time, to the BS. Based on this information, the BS can comprehensively consider and conduct UAV scheduling and planning.

First, assume that the UAV has a unified antenna array, and simultaneously sends an information signal to a K terrestrial communication user, represented by $\mathcal{K} = \{1, \ldots, K\}$. The K users has N calculation tasks in this system, represented by $\mathcal{N} = \{1, \ldots, N\}$. Each of these tasks is assigned at random. Each UAV has its communication link, and every task is offloaded onto the UAV and delivered to the user via this link. Consider a 3D (3D) Cartesian coordinate system in which each user's location is fixed at $(g_k, 0)$, in which $g_k = (x_k, y_k)$ is horizontal, $\boldsymbol{q}[t] = (x[t], y[t])$ is a time-varying location of a UAV at a time slot t, in which $(x[t], y[t])$ is a projection of a UAV. Accordingly, the channel gain at slot t between UAV node m and GU k is expressed as

$$h_{m,k}(\mathbf{q}[t], g_k) = \sqrt{\beta d^{-2}(\mathbf{q}[t], g_k)|} = \sqrt{\frac{\beta}{H_m^2 + \left\|\boldsymbol{q}[t] - g_k\right\|^2}}, \tag{1}$$

where β represents the channel power gain at the reference distance $d = 1$ m. $d(\boldsymbol{q}[t], g_k) = \sqrt{H_m^2 + \| \boldsymbol{q}[t] - g_k \|^2}$ represents the distance between the UAV and user k at slot t.

2.1. Communication model

Based on the above description, the signal interference plus noise ratio (SINR) of the GU k related to UAV node m is

$$SINR_{m,k} = \frac{p_T h_{m,k}}{p_G + B\sigma^2}, \tag{2}$$

where $p_G = \sum_{k' \neq k}^{K} p_T h_{m,k'} \omega(m, k')$ represents the sum of the interference power of other sub-channels. The σ^2 is the Gaussian noise of the channel, p_T is the signal transmission power from the GU to the UAV node, and B is the bandwidth of the UAV node in the air–ground communication link. $\omega(m, k')$ indicates whether the GU is associated with other UAVs, if it is associated, $\omega(m, k') = 1$, otherwise $\omega(m, k') = 0$. The transmission rate between the UAV node m and the GU k is given by

$$R_{m,k} = B \log_2(1 + SINR_{m,k}). \tag{3}$$

2.2. Sensing model

The UAV uses downlink NOMA technology to send signals $s = \sum_{k=1}^{K} \boldsymbol{w_k} s_k$ to k different communication users, where s_k is the data stream required by the kth user, and $\boldsymbol{w_k}$ is the beamforming vector formed corresponding to the kth data stream. In an integrated communication and sensing system, the waveform carrying communication should have the function of sense at the same time, so the covariance matrix of the transmitted signal can be designed as $\boldsymbol{R}_w = \sum_{k \in K} \boldsymbol{w_k} \boldsymbol{w_k^H}$. The function of sensing a target is to use the prior information of the target to maximize the effective sensed power, that is, the power of the detection signal in the direction of the target is given by

$$P(\theta_k) = \boldsymbol{a}^{\mathrm{H}}(\theta_k) \boldsymbol{R}_w \boldsymbol{a}(\theta_k), \tag{4}$$

where θ_k represents the azimuth angle of the UAV to the kth sensed target and denoted by

$$\boldsymbol{a}(\theta_k) = \left[1, \mathrm{e}^{\mathrm{j}2\pi \frac{d}{\lambda} \cos\theta_k}, \ldots, \mathrm{e}^{\mathrm{j}2\pi \frac{d}{\lambda}(E-1)\cos\theta_k}\right]^{\mathrm{T}}, \tag{5}$$

where $\boldsymbol{a}\left(\theta_k\right)$ is used to describe the response of a uniform array of E elements in the direction θ_k. λ and d are the carrier wavelength and the spacing between the two antennas, respectively.

The threshold value plays a crucial role in the sensing model. It serves as a benchmark for comparing the calculated effective sensing power. We define $P_{th} = k\sigma^2$, where k is a constant determined by the Neyman–Pearson criterion. In a typical radar system, $k = 3$. Accordingly, we set P_{th} to −85.2 dBm [31]. When the effective sensing power — maximized through the design of the transmitted signal's covariance matrix and the utilization of target prior information — reaches or exceeds this threshold, it signifies that the sensing operation has attained the required strength and accuracy. This implies that the system has successfully detected and recognized the target within the context of the application scenario.

2.3. Computing model

The information for task n contains the tuple (ζ_n, ψ_n, ξ_n), where ζ_n denotes the amount of input data for the task n, and ξ_n is the nth task generated by the GU. $\xi_n \in \mathcal{N}$, each user creates its own task at random. The nth task, which belongs to user k, is represented by ξ_n^k, and the total of all users' jobs is no more than $\mathcal{N}$. ψ_n is the amount of time needed to complete the task, which is proportional to the size of the input data. Thus, $\psi_n = \kappa\zeta_n$, where κ denotes the CPU cycle needed to execute a single bit of task data. The UAV nodes are responsible for executing the task scheduling matrix $\mathbf{X}$. Each element $x_{mn} \in \mathbf{X}$, where $n \in \mathcal{N}$ and $m \in \mathcal{M}$, is a binary variable that indicates if the UAV node m is offloaded with task n. The task n is offloaded to the UAV node m if $x_{mn} = 1$; otherwise, $x_{mn} = 0$, indicating that the task is not offloaded to that node.

2.3.1. Local task execution

Tasks can be executed locally at GUs. Assume that task n is generated in the GU k, which has a computational capacity of f_k CPU cycles per second. The computation delay for executing task n locally at GU k is expressed as

$$t_n^k = \frac{\psi_n}{f_k} = \frac{\kappa\zeta_n}{f_k}. \tag{6}$$

The energy consumption of local computation tasks is

$$e_n^k = \mu_k t_n^k, \tag{7}$$

where $\mu_k = \gamma f_k^3$, and μ_k is the computation energy consumption per unit time of GU k. γ is the effective switching capacitance.

2.3.2. UAV task execution

The transmission delay for offloading task n is given by

$$t_n^{tran} = \frac{\zeta_n}{R_{m,k}}. \tag{8}$$

Assuming that the computing power of the UAV node m (computed in CPU cycles per second) is f_m, the computing delay of the task n executed on the UAV node m is

$$t_n^{comp} = \frac{\psi_n}{f_m} = \frac{\kappa\zeta_n}{f_m}. \tag{9}$$

We assume that the size of the computed result is very small, and therefore, the delay caused by the drone returning the result to the user is negligible [29]. The overall delay of offloading task n is

$$t_n^c = t_n^{tran} + t_n^{comp}. \tag{10}$$

Typically, the output data of the task is significantly smaller than the input data, so the transmission energy consumption of the task result can be safely ignored. The μ_m is the UAV node m and computed energy consumption per unit time, and $\mu_m = \gamma f_m^3$, where γ is the effective switching capacitor, and the power consumption of the UAV node m for offloading task n is

$$e_n^c = \mu_m t_n^{comp} + p_T t_n^{tran}, \tag{11}$$

where p_T is the transmission power of the GU to the UAV node.

Therefore, the power of task execution is

$$P_c = (1 - x_{mn})\frac{e_n^k}{t_n^k} + x_{mn}\frac{e_n^c}{t_n^c}. \tag{12}$$

2.4. UAV hover and flight cost model

On the basis of optimizing and dispatching strategy of UAV, the optimal deployment of UAV nodes should be optimized according to the task scheduling strategy. Further consideration should be given to the power of updating the location of UAV nodes. Let $\boldsymbol{q}_m$ denote the UAV node m's new location vector after the deployment optimization strategy decision. Let $\boldsymbol{q}_m^0$ denote the UAV node's original location vector of UAV node m. We assume that the UAV m keeps the same movement throughout the location update and flies at v_m. Therefore, the flight time t_m can be expressed as:

$$t_m = \frac{\|\boldsymbol{q}_m - \boldsymbol{q}_m^0\|}{v_m}. \tag{13}$$

The UAV node needs to remain hovering during task computation. Assuming that the flight power of UAV node m is $p\left(v_m\right)$ and the cost of flying power in T_s is given by

$$e_m = e_m^{fly} + e_m^{hover}, \tag{14}$$

where the e_m^{fly} represents the energy consumption of UAV node m during flight, expressed as

$$e_m^{fly} = t_m p(v_m). \tag{15}$$

Based on the flight power model of UAV, the relation between $p(v_m)$ and v_m is

$$p(v_m) = P_0(1 + \frac{3v_m^2}{U_{tip}^2}) + P_i(\sqrt{1 + \frac{v_m^4}{4v_0^2}} - \frac{v_m^2}{2v_0^2})^{\frac{1}{2}} + \frac{1}{2}d_0\rho s A v_m^3, \tag{16}$$

where P_0 and P_i are the blade profile power and induced power in the floating state, U_{tip} is the tip speed of the rotor blade, v_0 is the mean induced speed of the rotor in the floating state, d_0 and s are the body drag ratio and the rotor solid, ρ and A respectively represent the air density and rotor disk area [32].

The e_m^{hover} is the hover energy consumption, and

$$e_m^{hover} = p(0)(T_s - t_m). \tag{17}$$

where $p(0)$ is

$$p(0) = P_0 + P_i. \tag{18}$$

the T_s represents the total amount of flying time.

3. Problem formulation and transformation

In this section, we formulate the problem of minimizing the total power by jointly optimizing multiple variables: the offloading decision, UAV speed, UAV trajectory, and the azimuth between the UAV and GU. The original problem is then transformed into a solvable form, and a sub-optimal solution is derived.

3.1. Problem formulation

The problem of minimizing the total power consumption can be formulated as

$$\mathbf{P1}: \quad \min_{\mathbf{x},\mathbf{v},\mathbf{q},\boldsymbol{\theta}} P = P(\theta_k) + P_c + p(v_m) + p(0) \tag{19a}$$

$$\text{s.t.} \quad \mathbf{C1}: \quad \sum_{m=1}^{M} x_{mn} = 1, \forall m, n, \tag{19b}$$

$$\mathbf{C2}: \quad \sum_{i=1}^{N} \| \boldsymbol{w}_i \|^2 \leqslant P_{\max}, \tag{19c}$$

$$\mathbf{C3}: \quad e_m^{fly} < e_{\max}^{fly}, \forall m, \tag{19d}$$

$$\mathbf{C4}: \quad \sum_{n}^{N} \psi_n x_{mn} \leqslant c_m, \forall m, \tag{19e}$$

$$\mathbf{C5}: \quad x_{mn} \in \{0, 1\}, \forall m, n, \tag{19f}$$

$$\mathbf{C6}: \quad \| \boldsymbol{q}[n] - \boldsymbol{q}[n-1] \|_2 \leq v_{\max} t, \forall n \in \mathcal{N}, \tag{19g}$$

$$\mathbf{C7}: \quad q[0] = q_1, \tag{19h}$$

$$\mathbf{C8}: \quad 0 \leq \theta_k \leq \pi, \tag{19i}$$

where **C1** ensures that each task can be offloaded at most once. The variable x_{mn} indicates that the nth task is offloaded to the mth UAV. **C2** constrains the sum of the beamforming vectors not to exceed the UAV's maximum transmitting power. **C3** ensures that the flight energy consumption of the UAV remains within the limit, where $e_{\max}^{fly}$ is the maximum allowable flight energy for the UAV. **C4** guarantees that the total offloaded tasks received by the UAV do not exceed its available computing resources, where c_m represents the computing capacity of the UAV node m. **C5** imposes binary constraints on x_{mn}. **C6** limits the maximum flight speed of the UAV; **C7** sets constraints based on the initial position of the UAV. **C8** ensures a range of angles between the UAV and the GU.

The optimization problem **P1** is a non-convex optimization problem. To address it, we decouple **P1** into three subproblems: the optimal offloading decision problem, the optimal UAV trajectory subproblem, and the optimal orientation angle subproblem.

3.2. Problem transformation

3.2.1. Offloading decision subproblem

When the UAV's position, the UAV's speed, and the UAV's angle to the user are determined, the problem is transformed into an offloading decision problem.

$$\mathbf{P2}: \quad \min_{\mathbf{x}} \sum_{n=1}^{N}\sum_{m=1}^{M}(1 - x_{mn})\frac{e_n^k}{t_n^k} + x_{mn}\frac{e_n^c}{t_n^c}, \tag{20a}$$

s.t. **C1**, **C4**, **C5**. (20b)

Based on the description of the system model, when UAV nodes handle different tasks with varying energy consumption and computational capacities, conventional offloading decision methods become highly inefficient. To address this, a multi-task offloading decision algorithm is required. We propose a task scheduling algorithm based on exchange matching, which ensures stable task allocation while also offering the advantages of low complexity and fast convergence.

We assume that the UAV is defined as $\mathcal{M} = \{1, \dots, M+1\}$ and that the task is set to $\mathcal{N} = \{1, \dots, N\}$, both of which are finite and disjoint. A matching pair (m, n) is created when task n is allocated to the UAV node m. The system power is used to build a preference list, which records the system power that one entity needs when it is matched with another in a swap. The matching exchange between the two entities is carried out in accordance with their respective priority lists. In this subproblem, every UAV node possesses a priority list for each task. Likewise, each task has its priority list for every UAV node. If the UAV node m_1 is ranked higher than the UAV node m_2 on the priority list of task n, it indicates that the total power required to offload task n to UAV node m_1 is less than that required by UAV node m_2. The priority lists of these UAV nodes are updated iteratively according to the following conditions. This algorithm ensures that no UAV fails to be assigned a task; in other words, each UAV will be matched with at least one task. Additionally, the preference among UAVs is transitive. For instance, based on the priority list of task n, UAV 1 ranks higher than UAV 2, and UAV 2 ranks higher than UAV 3, indicating that UAV 1 ranks higher than UAV 3.

We define a function that turns the offloading decision into a multi-match problem and it is described as follows.

$$| \mathcal{L}(n) | = 1, \forall n \in \mathcal{N}, \tag{21}$$

$$| \mathcal{L}(m) | \leq N_{\max}, \forall m \in \mathcal{M}, \tag{22}$$

$$\mathcal{L}(n) = m \Leftrightarrow \mathcal{L}(m) = n. \tag{23}$$

When tasks communicate with each other during scheduling, the state of one task can affect the performance of other tasks. This interdependence complicates task scheduling, especially when the matching of multiple task pairs needs to be considered. In this interference case, the utility of a task–resource matching pair may be affected by the presence of other task–resource matching pairs. This dependency renders the traditional definition of stability no longer applicable since traditional stability usually assumes that matching pairs are independent of each other. Therefore, we introduce the concept of exchange matching to optimize the overall matching returns by exchanging task and resource pairs. The proposed exchange matching algorithm begins by initializing task-UAV preferences using a delayed acceptance mechanism. Each task sequentially proposes to its most preferred UAV, which accepts or rejects based on its available resource capacity. In the exchange phase, the algorithm identifies task pairs (n, n') and UAV pairs (m, m') where swapping their assignments minimizes total power consumption. This iterative process continues until no further improvements can be made, resulting in a stable Nash equilibrium. We define an exchange operation so that the overall return can be improved by the exchange operation while keeping the partial matching pairs unchanged, as shown:

$$\mathcal{L}_{mn}^{m'n'} = \{\mathcal{L} \backslash \{(m,n),(m',n')\} \cup \{(m,n'),(m',n)\}\}. \tag{24}$$

Based on the problem **P2**, the preference cost function is defined as

$$V(m,n) = \begin{cases} \frac{e_n^l}{t_n^k}, x_{mn} = 0, \\ \frac{\mu_m t_n^{comp} + p_T t_n^{tran}}{t_n^{tran} + t_n^{comp}}, x_{mn} \neq 0. \end{cases} \tag{25}$$

At first, interference between GUs is not considered. In this paper, we employ the delayed acceptance algorithm (DAA) to create the initial matching between UAV nodes and tasks. Each task sends a request to the UAV node with the highest priority in its priority list, and every UAV node receives the task with the highest priority. The DAA algorithm ends when all the UAV nodes are fully used or when all the UAV nodes refuse to match the task.

Algorithm 1 Task Offloading Decision Algorithm Based On Exchange Matching

Input: Initialize the preference list A and B for UAVs and tasks, the matching state S_0 between the UAV node and the user, and set the number of iterations t to 1.
repeat
  for $(n \in \{1, ..., N\})$ **do**
    if A task $n' \in N(n' \neq n)$ can be found to make the powerless if the conditions are met, then swap n and n' according to preference list A, B **then**
      Swap (n, n') ;
    else
      Keep the current match status.
    end if
    Update the preference list according to the (25).
  end for
until Find the task offloading decision with the least power.
Output: Task offloading decision **x**.

In Algorithm 1, the preferred list of UAVs and tasks is first initialized, and then for the UAV node that has assigned tasks, the priority list is according to (25). According to the initialized matching status, for any task n, a swap will take place if there is another task $n' \in N(n' \neq n)$ that leads to a decrease in power consumption. The process continues until the system power stabilizes at a fixed value, and then the final result is written into the task scheduling matrix **x** for output.

The complexity analysis of Algorithm 1 is presented as follows. The priority queue is first initialized, and For M UAVs, the complexity of building the priority queue is $O(M log M)$. During the matching process, each UAV is popped from the priority queue with a complexity of $O(\log M)$. Therefore, the worst-case time complexity is $O(M \cdot N \log M)$. Thus, the overall complexity of the algorithm is $O(M \log M + M \cdot N \log M) = O(M \cdot N \log M)$. This complexity is lower than that of other comparable algorithms, enabling better performance.

3.2.2. UAV trajectory subproblem

When the offloading decision is finalized, the execution time t_n^{comp} for task n becomes independent of the deployment of the UAV. Thus, this sub-problem is equivalent to the optimization of the minimum value of communication power and flying power when the location and velocity of the UAV node change, as given by

$$\mathbf{P3}: \quad \min_{\mathbf{q},\mathbf{v},\theta} \sum_{n=1}^{N} \sum_{m=1}^{M} p_T x_{mn} + P(\theta_k) + p(v_m) + p(0), \tag{26a}$$

s.t. **C2**, **C3**, **C6**, **C7**, **C8**. (26b)

When the offloading decision variable x_{mn} is given, the UAV node location and velocity can be considered separately. First, problem **P3** is a non-convex, although the constraint **C6** is linear. However, since **P3** contains multiple second-order terms of $\|\vec{q_m} - \vec{q_k}\|^2$ and cannot be simplified directly, we employ the block coordinate descent method to handle q_m and v_m independently. Through this decomposition, we can reformulate **C3** as follows

$$e_m^{fly} - e_{\max}^{fly} < 0. \tag{27}$$

Lemma 1. $\varphi_m(x_m, y_m, H, v_m) = e_m^{fly} - e_{\max}^{fly}$ *is a convex function.*

Proof. See Appendix A. □

Algorithm 2 SCA-based UAV Trajectory Algorithm

Input: The position $\mathbf{q}$ and speed $\mathbf{v}$ of the UAV are initialized according to the task offloading decision matrix $\mathbf{x}$, and then the number of iterations $s = 0$.
repeat
  for $(m \in \{1, ..., M\})$ **do**
    for $(i \in \{1, ..., 3\})$ **do**
      Set t = 0, $\overline{I}_i^{s+1,0} = \overline{I}_i^{s}$.
      repeat
        $t = t + 1$;
        Using MATLAB convex problem solver to solve P4;
        Update $\overline{I}_i^{s+1,t}$ according to equation(29);
      until $\overline{I}_i^{s+1,t}$ is a stable convergent solution.
    end for
    $s = s + 1$.
  end for
until $\overline{I}^s$ is a stable convergent solution.
Output: The position $\mathbf{q}$ and speed $\mathbf{v}$ of the UAV, the azimuth angle between the UAV and the user $\theta_{\mathbf{k}}$.

To tackle the non-convexity issue, we employ the SCA algorithm to solve the optimization problem. Additionally, a multi-UAV trajectory planning algorithm is developed based on the SCA framework. We present the following lemma to support our approach:

Lemma 2. *The convex function $h(a; a^t)$ for any continuous differentiable first order nonconvex function $\Theta(a^t)$, there is*

$$h(a; a^t) = \Theta(a^t) + (\nabla\Theta(a^t) + C_1\vec{d})(a - a^t) + C_2\|\vec{b}(a - a^t)\|^2, \tag{28}$$

where C_1 and C_2 are positive constants, $\vec{b}$ and $\vec{d}$ are appropriate vectors, and t represents the iteration number, the iteration process is as follows:

$$a_{t+1} = a_t - \eta\left[\nabla\Theta(a_t) + 2\lambda(a_t - a_{t+1})\right], \tag{29}$$

where η is the learning rate.

According to the above lemma, we optimize the position variables (x_m, y_m, H) and velocity variables v_m, and then combine them into a new vector variable $\vec{I} = (x_m, y_m, H, v_m)$. Let $\vec{I}_i$ represent the ith element of $\vec{I}$. According to (28), the optimization problem can be converted to

$$\mathbf{P4}: \quad \min_{\vec{I}_i} h(I_i) = \Theta(I_i^t) + (\nabla\Theta(I_i^t) + C_1\vec{d})(I_i - I_i^t) + C_2\|\vec{b}(I_i - I_i^t)\|^2, \tag{30a}$$

$$\text{s.t.} \quad \mathbf{C2}, \mathbf{C3}, \mathbf{C6}, \mathbf{C7}, \mathbf{C8}, \tag{30b}$$

where I_i^t denotes the feasible point provided in the tth iteration. The standard convex optimization tool CVX can be employed to solve convex optimization problems. Algorithm 2 summarizes the UAV trajectory design algorithm based on the SCA approach. Once the UAV trajectory is determined, the power of the sensing model can be obtained by optimizing the angle between the GU and the UAV based on (4) in the sensing model.

The following is the algorithm complexity analysis of Algorithm 2. The time complexity of input initialization is $O(1)$, the outer loop is run M times, and each UAV is processed, so the time complexity is $O(M)$; the inner loop can be regarded as $O(1)$ for three times, and the problem is solved by the convex problem solver in MATLAB with the complexity of $O(n^3)$, where n is the number of variables, assuming that the number of stability check cycles is T, that is, the total time complexity of Algorithm 2 is $O(M \cdot T \cdot n^3)$. Compared with the exhaustive search algorithm, this algorithm achieves better performance.

We used the BCD algorithm to decompose problem P1 into two sub-problems and proposed the above two algorithms to solve the two problems respectively. Now we proposed a joint optimization Algorithm 3, which first initialized the offloading decision matrix $\mathbf{x_0}$, the initial position $\mathbf{q_0}$ of the UAV, and the initial speed $\mathbf{v_0}$ of the UAV, and then set a convergence accuracy ε. After entering the iteration, the offloading decision matrix $\mathbf{x}$ is updated by Algorithm 1, and the velocity matrix $\mathbf{v}$ and position matrix $\mathbf{q}$ of the UAV are updated by Algorithm 2, and then the cycle is continued until the convergence accuracy ε is reached. The time complexity analysis of Algorithm 3 is as follows. Because the time complexity of Algorithm 1 and Algorithm 2 is $O(M \cdot N \log M)$ and $O(M \cdot T \cdot n^3)$ respectively, the time complexity of Algorithm 3 can be obtained as $O(d(M \cdot N \log M + M \cdot T \cdot n^3))$ according to the process of Algorithm 3.

Algorithm 3 Proposed Joint Trajectory Design and Offloading Decision Optimization Algorithm

Input: Initialize the offloading decision matrix $\mathbf{x_0}$, the initial position $\mathbf{q_0}$ of the UAV trajectory, and the initial velocity $\mathbf{v_0}$ of the UAV, the number of iterations $d = 0$, and the convergence accuracy ε.
repeat
  $d = d + 1$;
  Update the offloading decision matrix $\mathbf{x}$ according to the offloading decision algorithm based on exchange matching in Algorithm 1;
  Update the UAV velocity matrix $\mathbf{v}$ and UAV position $\mathbf{q}$ matrix according to Algorithm 2;
until Achieve convergence accuracy ε
Output: The offloading decision matrix $\mathbf{x}$, the initial position $\mathbf{q}$ of the UAV trajectory, the initial velocity $\mathbf{v}$ of the UAV, the azimuth angle θ between the UAV and the user.

Table 1
Simulation parameters.

Parameter	Value	Parameter	Value
κ	1000	γ	10^{-27}
ζ_n	5~10 Mbits	v_0	5
A	0.79	f_k	10
U_{tip}	20	s	1
P_0	100	ρ	1.225
d_0	0.1	f_m	30~70
P_i	0.1	N	3
σ^2	−90 dBm	P_{th}	−85.2 dBm

4. Simulation results

In this section, we simulate the proposed offloading decision algorithm based on exchange matching, along with the SCA-based UAV trajectory algorithm, to achieve minimum power consumption. The simulation considers a scenario within a square area of 1.4 km by 1.4 km, involving three UAVs and four users, each assigned a single task. The detailed simulation parameters are provided in Table 1.

Fig. 2 illustrates the trend of computing power concerning the total task volume, comparing the proposed exchange matching-based offloading decision algorithm with the genetic and greedy algorithms. As the volume of tasks increases, the computing power required by all three algorithms also grows, following an expected trend. The figure clearly shows that the exchange matching-based algorithm outperforms the genetic and greedy algorithms in terms of computing power, consistently demonstrating lower power consumption. This indicates that selecting an algorithm with reduced complexity can significantly lower power consumption, thereby improving the efficiency of UAV-based computing systems.

Fig. 3 illustrates the trend of total model power consumption under different algorithms as UAV computing power varies. It can be seen from the figure that as UAV computing power increases, the total model power also grows across different algorithms. This increase occurs

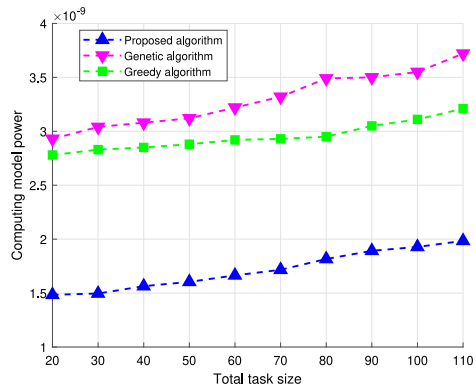

Fig. 2. Comparison of computing power consumption among the proposed exchange matching-based algorithm, genetic algorithm, and greedy algorithm with varying total task volumes.

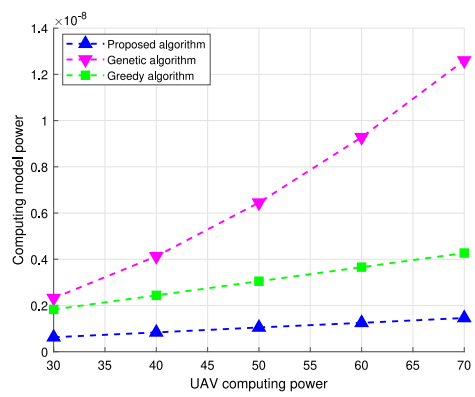

Fig. 3. Trend of total model power consumption under different algorithms as UAV computing power increases.

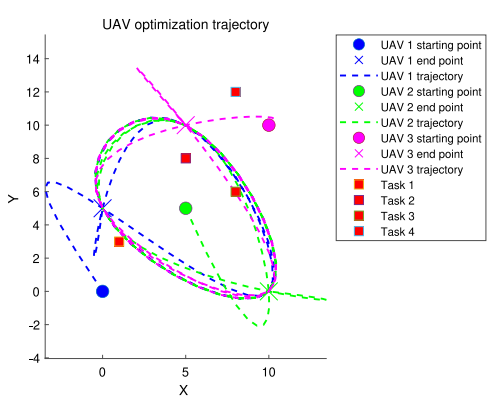

Fig. 4. Top view of UAV flight trajectories optimized using the SCA algorithm to minimize power consumption.

because enhancing computing capacity often requires UAVs to handle multiple tasks in parallel, leading to higher processor loads and energy consumption. Notably, the proposed algorithm achieves lower power consumption and a more gradual growth trend compared to the greedy and genetic algorithms. This indicates that, without compromising computing performance, power consumption can be further reduced by leveraging low-power processors, optimized software algorithms, and efficient cooling solutions.

Fig. 4 presents the top view of the flight trajectories for three UAVs operating within a 14 × 14 km^2 region, with user positions now included to provide a clearer understanding of the trajectory

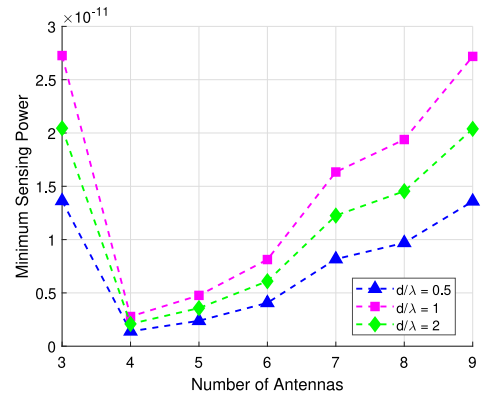

Fig. 5. Minimum sensing power versus the number of antennas under varying antenna spacing-to-wavelength ratios (d/λ).

optimization process. The trajectories are optimized using the SCA algorithm, aiming to minimize the overall power consumption while ensuring stable communication links and efficient task offloading. In this simulation, each UAV is optimized for a specific flight path based on user positions and task demands, dynamically adjusting its trajectory to serve ground users while maintaining efficient communication, sensing, and computing capabilities. The results demonstrate that the proposed method effectively reduces power consumption by optimizing the UAV trajectories to balance task allocation, communication quality, and energy efficiency. Additionally, the optimized paths show smooth and stable flight patterns, ensuring energy efficiency and operational reliability, validating the adaptability of the UAVs to real-time user demands. The simulation validates the capability of the SCA-based trajectory design to achieve near-minimum power consumption under the given constraints, demonstrating its effectiveness in practical multi-UAV network scenarios.

Fig. 5 illustrates the relationship between the minimum sensing power and the number of antennas at different antenna spacing-to-wavelength ratios (d/λ). The vertical axis represents the minimum sensing power, while the horizontal axis denotes the number of antennas, ranging from 3 to 9. The plot includes three distinct curves, each corresponding to a specific d/λ ratio. As observed from the figure, the minimum sensing power is achieved when the number of antennas is 4, with an antenna spacing-to-wavelength ratio of 0.5. The results suggest that increasing the number of antennas does not always lead to better performance, highlighting the importance of optimizing antenna configurations for power efficiency.

Fig. 6 shows the relationship between the UAV's flight power and speed at different initial speeds in the context of speed optimization. It is apparent that flight power reaches its minimum when the initial speed is 5 and the UAV's flight speed is 4. This indicates that, under these conditions, the balance between air resistance and energy consumption achieves an optimal level. Furthermore, the combined effects of the powertrain and air resistance at lower speeds can lead to increased power consumption.

Fig. 7 illustrates the relationship between the number of antennas and the total power consumption at varying ratios of antenna spacing to carrier wavelength. In comparison to Fig. 5, it is evident that when the number of antennas is set to 4, the power consumption is significantly lower at an antenna distance to carrier wavelength ratio of 0.5 than at other ratios. This observation highlights the importance of optimizing antenna configurations to achieve efficient power consumption, which is critical in the design of energy-efficient wireless communication systems.

Fig. 8 describes the relationship between the total task size and the total power of the offloading decision algorithm based on exchange matching proposed in this paper, compared with the greedy algorithm

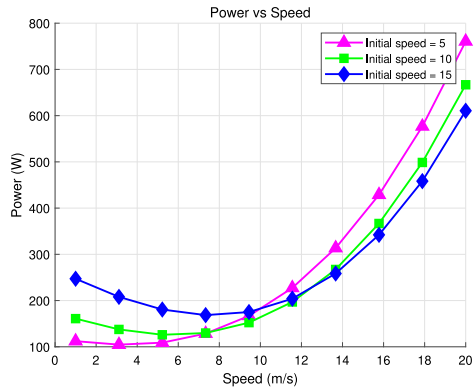

Fig. 6. Variation of UAV flight power at different initial speeds.

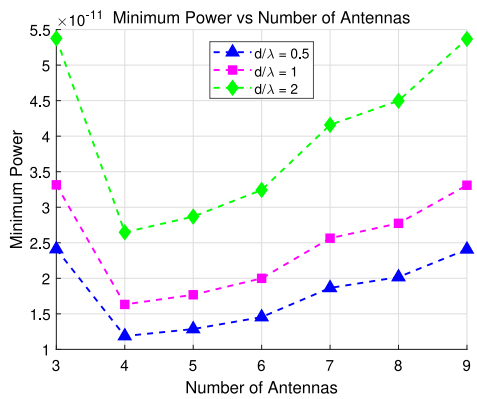

Fig. 7. Minimum total power versus the number of antennas under varying antenna spacing-to-wavelength ratios (d/λ).

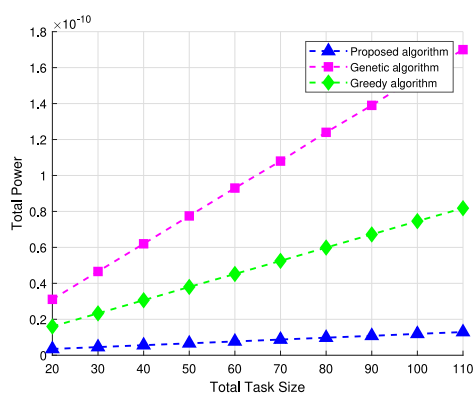

Fig. 8. Total power consumption of the proposed exchange matching algorithm compared to greedy and genetic algorithms across varying task sizes.

and genetic algorithm. It can be seen from the figure that when the total task size increases, the power consumption for all algorithms rises. Nonetheless, the algorithm proposed in this paper is far smaller than the other two algorithms, which once again confirms the superiority and advanced efficiency of the proposed algorithm in managing power consumption effectively.

Fig. 9 shows the relationship between the computing capacity of UAVs and the total power consumption under various algorithms. The offloading decision made by each algorithm is also closely tied to the total power. With the increase in computing power, the total power also increases. Compared with the greedy algorithm and genetic algorithm, our proposed algorithm consistently results in lower total

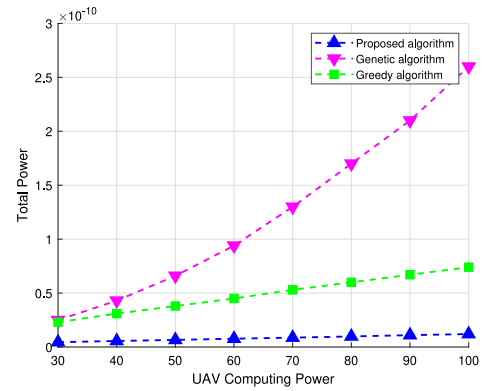

Fig. 9. Total power consumption concerning UAV computing capacity across different algorithms.

power consumption, further demonstrating the advanced efficiency of our approach.

5. Conclusion

This paper investigates the problem of UAV trajectory design and power minimization by leveraging communication, sensing, and computing technologies. We address this challenge by formulating a comprehensive optimization problem that involves four key variables: the UAV's trajectory, task offloading decisions, UAV speed, and the azimuth relative to the GU. We decompose the problem into two sub-problems. The first sub-problem is the offloading decision, for which we propose an algorithm based on exchange matching. The second sub-problem addresses UAV trajectory optimization. Given its non-convex nature, we introduce Lemmas 1 and 2 to convert it into a convex problem, enabling the use of a convex problem solver. Our SCA-based UAV trajectory algorithm effectively minimizes power consumption during both UAV hovering and in-flight operations. We achieve joint optimization by combining these two algorithms using the BCD framework. Simulation results demonstrate that our proposed joint optimization algorithm significantly outperforms existing benchmark algorithms, underscoring its effectiveness and potential in reducing power consumption for UAV operations.

Furthermore, while this study primarily optimizes UAV trajectory and power consumption, future research could explore more dynamic task-handling aspects, such as task dependencies, varying computational demands, and real-time arrivals. Additionally, leveraging advanced learning-based techniques like deep reinforcement learning may enhance UAV adaptability in complex environments. Another key direction involves validating our approach in real-world UAV-IoT deployments, considering challenges such as unpredictable user mobility, fluctuating channel conditions, and environmental constraints. These advancements would not only improve the robustness of our method but also contribute to the development of more intelligent and autonomous UAV-assisted networks.

CRediT authorship contribution statement

Zhongyu Wang: Writing – review & editing, Writing – original draft, Validation, Resources, Methodology, Investigation. **Hongbo Meng:** Writing – review & editing, Writing – original draft, Validation, Resources. **Jihang Shi:** Writing – original draft. **Jiaen Zhou:** Writing – original draft. **Yashuai Cao:** Writing – original draft. **Guanghua Gu:** Writing – original draft. **Xuehua Li:** Writing – original draft.

Declaration of competing interest

The authors have no conflict of interest to declare.

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Appendix A. Proof of Lemma 1

Proof. Since $e_{\max}^{fly}$ is a constant, it only needs to prove that e_m^{fly} is a convex function. Therefore, rewrite e_m^{fly} as:

$$e_m^{fly} = t_m p_m(v_m) = \phi(x_m, y_m, H)\delta(v_m), \tag{A.1}$$

where

$$\phi(x_m, y_m, H) = \left\|\overrightarrow{q_m} - \overrightarrow{q_m}^0\right\| = \sqrt{(x_m - x_m^0)^2 + (y_m - y_m^0)^2 + H^2}, \tag{A.2}$$

$$\delta(v_m) = \frac{p_m}{v_m} = \frac{P_0}{v_m}\left(1 + \frac{3v_m^2}{U_{tip}^2}\right) + \frac{P_i}{v_m}\left(\sqrt{1 + \frac{v_m^4}{4v_0^2}} - \frac{v_m^2}{2v_0^2}\right)^{\frac{1}{2}} + \frac{1}{2}d_0\rho s A v_m^2. \tag{A.3}$$

We compute the second-order partial derivatives of each of the four variables within the aforementioned equation. When only the position variables (x_m, y_m, H) are considered, we simply approximate the formula as $e_m^{fly} = C_1\phi(x_m, y_m, H)$, where C_1 is a non-negative coefficient that has no impact on the bump of the problem. Easy to know, when only one position variable is considered, $\phi(x_m, y_m, H)$ is a convex function.

When we focus solely on the velocity variable v_m, the problem simplifies to $e_m^{fly} = C_2\delta(v_m)$. Here, C_2 is a non-negative coefficient that has no impact on the essence of the problem. By computing the second partial derivative of v_m in $\delta(v_m)$, we can get that the second partial derivative is always greater than 0, so φ_m is a convex function.

The proof of Lemma 1 is completed. ■

Appendix B. Proof of BCD algorithm

Proof. Let the original problem **P1** be defined as:

$$\min_{(\mathbf{x},\mathbf{q},\mathbf{v},\theta)} P(\mathbf{x}, \mathbf{q}, \mathbf{v}, \theta) \tag{B.1}$$

subject to constraints (19b)–(19i).

The BCD method iteratively optimizes subsets of variables while fixing others. At each iteration t: subproblem **P2** fixes $\mathbf{q}^t, \mathbf{v}^t, \theta^t$, and solves for $\mathbf{x}^{t+1}$. Subproblem **P3** fixes $\mathbf{x}^{t+1}$, and solves for $\mathbf{q}^{t+1}, \mathbf{v}^{t+1}, \theta^{t+1}$.

To show equivalence, we prove that the KKT conditions of **P1** are satisfied if and only if the KKT conditions of **P2** and **P3** are satisfied at convergence:

For **P2** : Fixing $\mathbf{q}, \mathbf{v}, \theta$ the KKT conditions reduce to $\nabla_{\mathbf{x}} P + \lambda\nabla_{\mathbf{x}}(C1, C4, C5) = 0$.

For **P3**: Fixing $\mathbf{x}$, the KKT conditions reduce to $\nabla_{\mathbf{q},\mathbf{v},\theta} P + \mu\nabla_{\mathbf{q},\mathbf{v},\theta}(C2, C3, C6 - C8) = 0$.

Since BCD alternates between satisfying these partial KKT conditions, the solution at convergence satisfies the full KKT conditions of **P1**.

Appendix C. Convergence of BCD algorithm

The BCD algorithm guarantees monotonic convergence to a stationary point of **P1**.

Proof. Let $P^t = P(\mathbf{x^t}, \mathbf{q^t}, \mathbf{v^t}, \theta^t)$ subject to constraints (19b)–(19i).

Step 1 (Solve **P2**): Minimizing the total power consumption P over $\mathbf{x}$:

$$P^{(t+1)/2} = \min_{\mathbf{x}} P(\mathbf{x^t}, \mathbf{q^t}, \mathbf{v^t}, \theta^t) \le P^t. \tag{C.1}$$

Step 2 (Solve **P3**): Minimizing the total power consumption P over $\mathbf{q}, \mathbf{v}, \theta$:

$$P^{t+1} = \min_{\mathbf{q},\mathbf{v},\theta} P(\mathbf{x^{t+1}}, \mathbf{q}, \mathbf{v}, \theta) \le P^{(t+1)/2}. \tag{C.2}$$

Thus, $P^{t+1} \le P^t$ and since P is bounded below (energy cannot be negative), the sequence P^t converges.

Data availability

The data that has been used is confidential.

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